Intro to Rational Expressions – Fractions and Exponents Review – Worksheet

MCR3U

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SOLUTIONS

1) Add or subtract the following fractions.

a) $\frac{2}{3}+\frac{3}{4}$

$= \frac{8}{12}+\frac{9}{12}$

$= \frac{17}{12}$

b) $\frac{3}{5}-\frac{1}{3}$

$= \frac{9}{15}-\frac{5}{15}$

$= \frac{4}{15}$

c) $\frac{x}{3}+\frac{y}{2}$

$= \frac{2x}{6}+\frac{3y}{6}$

$= \frac{2x+3y}{6}$

d) $\frac{3x}{5}-\frac{2y}{7}$

$= \frac{21x}{35}-\frac{10y}{35}$

$= \frac{21x-10y}{35}$

2) Simplify (multiplication)

a) $\left(\frac{2}{3}\right)\left(-\frac{1}{5}\right)$

$= \frac{-2}{15}$

b) $\left(\frac{\cancel{4}^{1}}{\cancel{5}_{1}}\right)\left(\frac{\cancel{25}^{5}}{\cancel{12}_{3}}\right)$

$= \frac{5}{3}$

c) $\left(\frac{17}{8}\right)\left(\sqrt{\frac{13}{289}}\right)$

$= \frac{17}{8}\left(\frac{\sqrt{13}}{17}\right)$

$= \frac{\sqrt{13}}{8}$

d) $\left(-\frac{5}{8}\right)\left(\sqrt{\frac{75}{128}}\right)$

$= \left(\frac{-5}{8}\right)\left(\frac{5\sqrt{3}}{8\sqrt{2}}\right)$

$= \frac{-25\sqrt{3}}{64\sqrt{2}}$

3) Simplify (multiplication and division)

a) $\frac{\cancel{4}}{9}\cdot\frac{7}{\cancel{4}}$

$= \frac{7}{9}$

b) $\frac{-11}{5}\cdot\frac{-7}{4}$

$= \frac{77}{20}$

c) $\frac{-17}{10}\div\frac{9}{4}$

$= \frac{-17}{5\cancel{10}_{2}}\times\frac{\cancel{4}^{2}}{9}$

$= \frac{-34}{45}$

d) $\frac{6}{7}\div\frac{7}{6}$

$= \frac{6}{7}\times\frac{6}{7}$

$= \frac{36}{49}$

e) $\frac{7}{\cancel{3}_{1}}\cdot\frac{\cancel{6}^{2}}{1}$

$= 14$

4) Simplify (exponents)

a) $(x^6)(x^4)$

$= x^{10}$

b) $(y^3)(y^7)(y^9)$

$= y^{19}$

c) $m^5 \div m^4$

$= m$

d) $h^6 \div h^8$

$= h^{-2}$

$= \frac{1}{h^2}$

e) $\sqrt{h^6} \div \sqrt{h^8}$

$= (h^6)^{\frac{1}{2}} \div (h^8)^{\frac{1}{2}}$

$= h^3 \div h^4$

$= \frac{1}{h}$

f) $\frac{x^{10}}{x^5}$

$= x^5$

g) $\left(yz^2\right)^3$

$= y^3 z^6$

h) $\left[-(x)^2\right]^2$

$= x^4$

i) $\left(x^2\right)^3\left(y^3\right)^2$

$= x^6 y^6$

j) $\frac{(-x)^2}{(-x)}$

$= \frac{x^2}{-x^1}$

$= -x$

5) Simplify fractions and exponents

a) $\frac{7x}{3y} \cdot \frac{12y}{5x^2} + \frac{2}{3}$

$\frac{28xy}{5x^2y} + \frac{2}{3}$

$\frac{28}{5x} + \frac{2}{3}$

$\frac{84}{15x} + \frac{10x}{15x}$

$= \frac{84+10x}{15x}$

b) $5\frac{25x^3}{3x} \cdot \frac{7y^2}{5yx^2}$

$= \frac{35x^3y^2}{3x^3y}$

$= \frac{35y}{3}$

6) Simplify. Your answer should contain only positive exponents.

a) $(x^{-2}x^{-3})^4$

$= (x^{-5})^4$

$= x^{-20}$

$= \frac{1}{x^{20}}$

b) $(x^4)^{-3} \cdot 2x^4$

$= x^{-12} \cdot 2x^4$

$= 2x^{-8}$

$= \frac{2}{x^8}$

c) $(n^3)^3 \cdot 2n^{-1}$

$= n^9 \cdot 2n^{-1}$

$= 2n^8$

d) $(2v)^2 \cdot 2v^2$

$= 4v^2 \cdot 2v^2$

$= 8v^4$

e) $\frac{2x^2y^4 \cdot 4x^2y^4 \cdot 3x}{3x^{-3}y^2}$

$= \frac{24x^5y^8}{3x^{-3}y^2}$

$= 8x^8y^6$

f) $\frac{2y^3 \cdot 3xy^3}{3x^2y^4}$

$= \frac{6xy^6}{3x^2y^4}$

$= \frac{2y^2}{x}$

g) $\frac{x^3y^3 \cdot x^3}{4x^2}$

$= \frac{x^6y^3}{4x^2}$

$= \frac{x^4y^3}{4}$

h) $\frac{3x^2y^2}{2x^{-1} \cdot 4yx^2}$

$= \frac{3x^2y^2}{8xy}$

$= \frac{3xy}{8}$

i) $\frac{x}{(2x^0)^2}$

$= \frac{x}{4}$

j) $\frac{2m^{-4}}{(2m^{-4})^3}$

$= \frac{2m^{-4}}{8m^{-12}}$

$= \frac{m^8}{4}$

Answers

1. a) $\frac{17}{12}$ **b)** $\frac{4}{15}$ **c)** $\frac{2x+3y}{6}$ **d)** $\frac{21x-10y}{35}$

2. a) $-\frac{2}{15}$ **b)** $\frac{5}{3}$ **c)** $\frac{\sqrt{13}}{8}$ **d)** $-\frac{25\sqrt{3}}{64\sqrt{2}}$

3. a) (7/9) **b)** (77/20) **c)** (-34/45) **d)** (36/49) **e)** 14

4. a) x^{10} **b)** y^{19} **c)** m **d)** ~~h^2~~ $\frac{1}{h^2}$ **e)** $1/h$ **f)** x^5 **g)** y^3z^6 **h)** x^4 **i)** x^6y^6 **j)** $-x$

5. a) $\frac{84+10x}{15x}$ **b)** $\frac{35y}{3}$

6) a) $\frac{1}{x^{20}}$ **b)** $\frac{2}{x^8}$ **c)** $2n^8$ **d)** $8v^4$ **e)** $8x^8y^6$ **f)** $\frac{2y^2}{x}$ **g)** $\frac{x^4y^3}{4}$ **h)** $\frac{3xy}{8}$ **i)** $\frac{x}{4}$ **j)** $\frac{m^8}{4}$